a. (1 mark) Some students took French in spring 2001.
$\exists x\ Student(x) \wedge Takes(x, F, Spring2001)$.
b. (1 mark) Every student who takes French passes it.
$\forall x, s\ Student(x) \wedge Takes(x, F, s) \Rightarrow Passes(x, F, s)$.
c. (1 mark) Only one student took Greek in spring 2001.
$\exists x\ Student(x) \wedge Takes(x,G, Spring2001) \wedge \forall y\ y \_= x \Rightarrow \neg Takes(y,G, Spring2001)$.
d. (1 mark) The best score in Greek is always higher than the best score in French.
$\forall s\ \exists x\ \forall y\ Score(x,G, s) > Score(y,F, s)$.
e. (1 mark) Every person who buys a policy is smart.
$\forall x\ Person(x) \wedge (\exists y, z\ Policy(y) \wedge Buys(x, y, z)) \Rightarrow Smart(x)$.
f. (1 mark) No person buys an expensive policy.
$\forall x, y, z\ Person(x) \wedge Policy(y) \wedge Expensive(y) \Rightarrow \neg Buys(x, y, z)$.
g. (1 mark) There is an agent who sells policies only to people who are not insured.
$\exists x\ Agent(x) \wedge \forall y, z\ Policy(y) \wedge Sells(x, y, z) \Rightarrow (Person(z) \wedge \neg Insured(z))$.
h. (1 mark) There is a barber who shaves all men in town who do not shave themselves.
$\exists x\ Barber(x) \wedge \forall y\ Man(y) \wedge \neg Shaves(y, y) \Rightarrow Shaves(x, y)$.
i. (2 marks) A person born in the UK, each of whose parents is a UK citizen or a UK resident, is a UK citizen by birth.
$\forall x\ Person(x) \wedge Born(x,UK) \wedge (\forall y\ Parent(y, x) \Rightarrow ((\exists r\ Citizen(y,UK, r)) \vee Resident(y,UK))) \Rightarrow Citizen(x,UK,Birth)$.
j. (2 marks) A person born outside the UK, one of whose parents is a UK citizen by birth, is a UK citizen by descent.
$\forall x\ Person(x) \wedge \neg Born(x,UK) \wedge (\exists y\ Parent(y, x) \wedge Citizen(y,UK,Birth)) \Rightarrow Citizen(x,UK,Descent)$.
k. (2 marks) Politicians can fool some of the people all of the time, and they can fool all of the people some of the time, but they can't fool all of the people all of the time.
$\forall x\ Politician(x) \Rightarrow (\exists y\ \forall t\ Person(y) \wedge Fools(x, y, t)) \wedge (\exists t\ \forall y\ Person(y) \Rightarrow Fools(x, y, t)) \wedge \neg(\forall t\ \forall y\ Person(y) \Rightarrow Fools(x, y, t))$

2. Consider the following axioms:

1. Every child loves Santa.
   $\forall x\ (CHILD(x) \rightarrow LOVES(x,Santa))$
2. Everyone who loves Santa loves any reindeer.
   $\forall x\ (LOVES(x,Santa) \rightarrow \forall y\ (REINDEER(y) \rightarrow LOVES(x,y)))$
3. Rudolph is a reindeer, and Rudolph has a red nose.
   $REINDEER(Rudolph) \wedge REDNOSE(Rudolph)$
4. Anything which has a red nose is weird or is a clown.
   $\forall x\ (REDNOSE(x) \rightarrow WEIRD(x) \vee CLOWN(x))$
5. No reindeer is a clown.
   $\neg \exists x\ (REINDEER(x) \wedge CLOWN(x))$
6. Scrooge does not love anything which is weird.
   $\forall x\ (WEIRD(x) \rightarrow \neg LOVES(Scrooge,x))$

7. (Conclusion) Scrooge is not a child.
   $\neg\ CHILD(Scrooge)$

3. Consider the following axioms:

1. Anyone who buys carrots by the bushel owns either a rabbit or a grocery store.
   $\forall x\ (BUY(x) \rightarrow \exists y\ (OWNS(x,y) \wedge (RABBIT(y) \vee GROCERY(y))))$
2. Every dog chases some rabbit.
   $\forall x\ (DOG(x) \rightarrow \exists y\ (RABBIT(y) \wedge CHASE(x,y)))$
3. Mary buys carrots by the bushel.
   $BUY(Mary)$
4. Anyone who owns a rabbit hates anything that chases any rabbit.
   $\forall x\ \forall y\ (OWNS(x,y) \wedge RABBIT(y) \rightarrow \forall z\ \forall w\ (RABBIT(w) \wedge CHASE(z,w) \rightarrow HATES(x,z)))$
5. John owns a dog.
   $\exists x\ (DOG(x) \wedge OWNS(John,x))$
6. Someone who hates something owned by another person will not date that person.
   $\forall x\ \forall y\ \forall z\ (OWNS(y,z) \wedge HATES(x,z) \rightarrow \neg\ DATE(x,y))$
7. (Conclusion) If Mary does not own a grocery store, she will not date John.
   $((\neg\ \exists x\ (GROCERY(x) \wedge OWN(Mary,x))) \rightarrow \neg\ DATE(Mary,John))$

4. Consider the following axioms:

1. Every Austinite who is not conservative loves some armadillo.
   $\forall x\ (AUSTINITE(x) \wedge \neg\ CONSERVATIVE(x) \rightarrow \exists y\ (ARMADILLO(y) \wedge LOVES(x,y)))$
2. Anyone who wears maroon-and-white shirts is an Aggie.
   $\forall x\ (WEARS(x) \rightarrow AGGIE(x))$
3. Every Aggie loves every dog.
   $\forall x\ (AGGIE(x) \rightarrow \forall y\ (DOG(y) \rightarrow LOVES(x,y)))$
4. Nobody who loves every dog loves any armadillo.
   $\neg\ \exists x\ ((\forall y\ (DOG(y) \rightarrow LOVES(x,y))) \wedge \exists z\ (ARMADILLO(z) \wedge LOVES(x,z)))$
5. Clem is an Austinite, and Clem wears maroon-and-white shirts.
   $AUSTINITE(Clem) \wedge WEARS(Clem)$
6. (Conclusion) Is there a conservative Austinite?
   $\exists x\ (AUSTINITE(x) \wedge CONSERVATIVE(x))$

```
( ( (not (Austinite x))  (Conservative x)  (Armadillo (f x)) )
  ( (not (Austinite x))  (Conservative x)  (Loves x (f x)) )
  ( (not (Wears x))  (Aggie x) )
  ( (not (Aggie x))  (not (Dog y))  (Loves x y) )
  ( (Dog (g x))  (not (Armadillo z))  (not (Loves x z)) )
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```
( (not (Loves x (g x)))  (not (Armadillo z))  (not (Loves x z)) )
( (Austinite (Clem)) )
( (Wears (Clem)) )
( (not (Conservative x))  (not (Austinite x)) ) )
```

5. Consider the following axioms:

1. Anyone whom Mary loves is a football star.
   $\forall x\ (LOVES(Mary,x) \rightarrow STAR(x))$
2. Any student who does not pass does not play.
   $\forall x\ (STUDENT(x) \wedge \neg\ PASS(x) \rightarrow \neg\ PLAY(x))$
3. John is a student.
   $STUDENT(John)$
4. Any student who does not study does not pass.
   $\forall x\ (STUDENT(x) \wedge \neg\ STUDY(x) \rightarrow \neg\ PASS(x))$
5. Anyone who does not play is not a football star.
   $\forall x\ (\neg\ PLAY(x) \rightarrow \neg\ STAR(x))$
6. (Conclusion) If John does not study, then Mary does not love John.
   $\neg\ STUDY(John) \rightarrow \neg\ LOVES(Mary,John)$

6. Consider the following axioms:

1. Every coyote chases some roadrunner.
   $\forall x\ (COYOTE(x) \rightarrow \exists y\ (RR(y) \wedge CHASE(x,y)))$
2. Every roadrunner who says ``beep-beep'' is smart.
   $\forall x\ (RR(x) \wedge BEEP(x) \rightarrow SMART(x))$
3. No coyote catches any smart roadrunner.
   $\neg\ \exists x\ \exists y\ (COYOTE(x) \wedge RR(y) \wedge SMART(y) \wedge CATCH(x,y))$
4. Any coyote who chases some roadrunner but does not catch it is frustrated.
   $\forall x\ (COYOTE(x) \wedge \exists y\ (RR(y) \wedge CHASE(x,y) \wedge \neg\ CATCH(x,y)) \rightarrow FRUSTRATED(x))$
5. (Conclusion) If all roadrunners say ``beep-beep'', then all coyotes are frustrated.
   $(\forall x\ (RR(x) \rightarrow BEEP(x)) \rightarrow (\forall y\ (COYOTE(y) \rightarrow FRUSTRATED(y)))$

```
( ( (not (Coyote x))  (RR (f x)) )
  ( (not (Coyote x))  (Chase x (f x)) )
  ( (not (RR x))  (not (Beep x))  (Smart x) )
  ( (not (Coyote x))  (not (RR y))  (not (Smart y))  (not (Catch x y)) )
  ( (not (Coyote x))  (not (RR y))  (not (Chase x y)) (Catch x y)
    (Frustrated x) )
  ( (not (RR x))  (Beep x) )
  ( (Coyote (a)) )
  ( (not (Frustrated (a))) ) )
```

7. Consider the following axioms:

1. Anyone who rides any Harley is a rough character.
   $\forall x\ ((\exists y\ (HARLEY(y) \wedge RIDES(x,y))) \rightarrow ROUGH(x))$
2. Every biker rides [something that is] either a Harley or a BMW.
   $\forall x\ (BIKER(x) \rightarrow \exists y\ ((HARLEY(y) \vee BMW(y)) \wedge RIDES(x,y)))$
3. Anyone who rides any BMW is a yuppie.
   $\forall x\ \forall y\ (RIDES(x,y) \wedge BMW(y) \rightarrow YUPPIE(x))$
4. Every yuppie is a lawyer.
   $\forall x\ (YUPPIE(x) \rightarrow LAWYER(x))$
5. Any nice girl does not date anyone who is a rough character.
   $\forall x\ \forall y\ (NICE(x) \wedge ROUGH(y) \rightarrow \neg\ DATE(x,y))$
6. Mary is a nice girl, and John is a biker.
   $NICE(Mary) \wedge BIKER(John)$
7. (Conclusion) If John is not a lawyer, then Mary does not date John.
   $\neg\ LAWYER(John) \rightarrow \neg\ DATE(Mary,John)$

8. Consider the following axioms:

1. Every child loves anyone who gives the child any present.
   $\forall x\ \forall y\ \forall z\ (CHILD(x) \wedge PRESENT(y) \wedge GIVE(z,y,x) \rightarrow LOVES(x,z)$
2. Every child will be given some present by Santa if Santa can travel on Christmas eve.
   $TRAVEL(Santa,Christmas) \rightarrow \forall x\ (CHILD(x) \rightarrow \exists y\ (PRESENT(y) \wedge GIVE(Santa,y,x)))$
3. It is foggy on Christmas eve.
   $FOGGY(Christmas)$
4. Anytime it is foggy, anyone can travel if he has some source of light.
   $\forall x\ \forall t\ (FOGGY(t) \rightarrow (\ \exists y\ (LIGHT(y) \wedge HAS(x,y)) \rightarrow TRAVEL(x,t)))$
5. Any reindeer with a red nose is a source of light.
   $\forall x\ (RNR(x) \rightarrow LIGHT(x))$
6. (Conclusion) If Santa has some reindeer with a red nose, then every child loves Santa.
   $(\ \exists x\ (RNR(x) \wedge HAS(Santa,x))) \rightarrow \forall y\ (CHILD(y) \rightarrow LOVES(y,Santa))$

9. Consider the following axioms:

1. Every investor bought [something that is] stocks or bonds.
   $\forall x\ (INVESTOR(x) \rightarrow \exists y\ ((STOCK(y) \vee BOND(y)) \wedge BUY(x,y)))$
2. If the Dow-Jones Average crashes, then all stocks that are not gold stocks fall.
   $DJCRASH \rightarrow \forall x\ ((STOCK(x) \wedge \neg\ GOLD(x)) \rightarrow FALL(x))$

3. If the T-Bill interest rate rises, then all bonds fall.
   $TBRISE \rightarrow \forall x\ (BOND(x) \rightarrow FALL(x))$
4. Every investor who bought something that falls is not happy.
   $\forall x\ \forall y\ (INVESTOR(x) \wedge BUY(x,y) \wedge FALL(y)\ \text{\&rarrm;}\ \neg\ HAPPY(x))$
5. (Conclusion) If the Dow-Jones Average crashes and the T-Bill interest rate rises, then any investor who is happy bought some gold stock.
   $(\ DJCRASH \wedge TBRISE\ ) \rightarrow \forall x\ (INVESTOR(x) \wedge HAPPY(x) \rightarrow \exists y\ (GOLD(y) \wedge BUY(x,y)))$

10. Consider the following axioms:

1. Every child loves every candy.
   $\forall x\ \forall y\ (CHILD(x) \wedge CANDY(y) \rightarrow LOVES(x,y))$
2. Anyone who loves some candy is not a nutrition fanatic.
   $\forall x\ (\ (\exists y\ (CANDY(y) \wedge LOVES(x,y))) \rightarrow \neg\ FANATIC(x))$
3. Anyone who eats any pumpkin is a nutrition fanatic.
   $\forall x\ ((\exists y\ (PUMPKIN(y) \wedge EAT(x,y))) \rightarrow FANATIC(x))$
4. Anyone who buys any pumpkin either carves it or eats it.
   $\forall x\ \forall y\ (PUMPKIN(y) \wedge BUY(x,y) \rightarrow CARVE(x,y) \vee EAT(x,y))$
5. John buys a pumpkin.
   $\exists x\ (PUMPKIN(x) \wedge BUY(John,x))$
6. Lifesavers is a candy.
   $CANDY(Lifesavers)$
7. (Conclusion) If John is a child, then John carves some pumpkin.
   $CHILD(John) \rightarrow \exists x\ (PUMPKIN(x) \wedge CARVE(John,x))$